(8) (i) real and distinct (ii) real and distinct (iii) real and distinct

(9) (i) not intersect (ii) intersects at two points (iii) touches at only one point

(10) $\left(x+\frac{5}{2}\right)^2-\left(\frac{3}{2}\right)^2$

Exercise 2.5

(1) $\left[-3,\frac{5}{2}\right]$ (2) $[1,2]$

Exercise 2.6

(1) $-\frac{5}{2}$ and $\frac{5}{2}$ (2) $\frac{3+\sqrt{53}}{2}$ and $\frac{3-\sqrt{53}}{2}$ (3) $x=\pm 2$ (4) $x=-\frac{3}{5}$ or -1

Exercise 2.7

(1) $(x^2+\sqrt{2}x+1)(x^2-\sqrt{2}x+1)$ (2) $a=5$

Exercise 2.8

(1) $(0,1)\cup(2,\infty)$ (2) $\left(-\infty,\frac{3}{2}\right)\cup(2,4)$ (3) $(-3,-2]\cup[2,5)$

Exercise 2.9

(1) $\frac{1}{2a(x-a)}-\frac{1}{2a(x+a)}$

(2) $\frac{7}{3(x-2)}+\frac{2}{3(x+1)}$

(3) $\frac{1}{6(x-1)}+\frac{2}{15(x+2)}+\frac{-3x+1}{10(x^2+1)}$

(4) $\frac{1}{(x-1)^2}+\frac{1}{(x-1)^3}$

(5) $\frac{1}{4(x-1)}-\frac{1}{4(x+1)}-\frac{1}{2(x^2+1)}$

(6) $\frac{1}{x}-\frac{2}{x^2+1}$

(7) $1+\frac{13}{x-3}-\frac{7}{x-2}$

(8) $(x-5)+\frac{32}{(x+3)}-\frac{11}{(x+2)}$

(9) $-\frac{14}{9(x+1)}-\frac{11}{3(x+1)^2}+\frac{14}{9(x-2)}$

(10) $\frac{4}{x+1}+\frac{2x-3}{x^2+1}$

(11) $2+\left(\frac{2}{x+3}-\frac{1}{x-1}\right)$

(12) $\frac{3}{x+1}+\frac{4-3x}{x^2+1}$

Exercise 2.11

(1)(i) 25 (ii) $\frac{1}{8}$ (iii) $\frac{1}{100}$ (iv) $\frac{1}{9}$ (v) $\frac{1}{3}$

(2) 8 (3) $\frac{1}{\sqrt{2}}$ (4) $\frac{1}{2}$ (5) 2